

Omega MS2

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Circuit Purpose

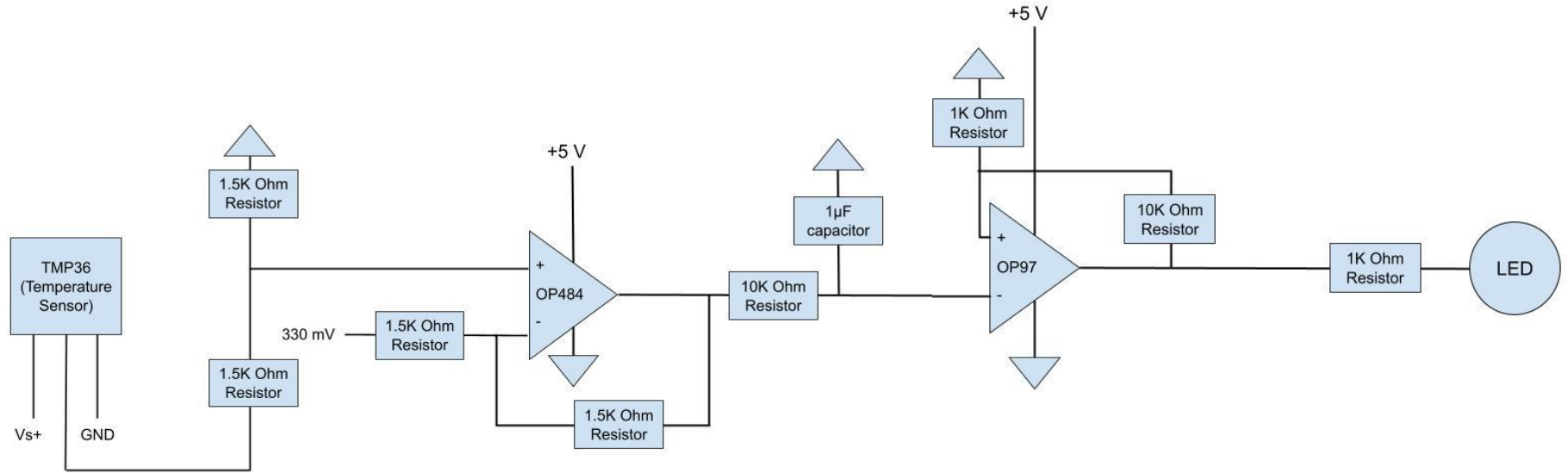
The circuit is designed to activate an output when a specific temperature is reached using the TMP36 temperature sensor. The TMP36 produces a voltage proportional to temperature, which is first amplified and conditioned using a differential amplifier by the OP484 to improve signal accuracy.

The signal then passes through an RC low-pass filter to remove noise and prevent unstable behavior. After filtering, the voltage is compared to a set reference threshold (0.33 V) using the OP97 configured as a comparator.

When the measured temperature voltage exceeds the reference voltage, the comparator outputs a high signal, turning on the LED. When the temperature is below the threshold, the output remains low and the LED stays off.

This allows the circuit to provide a clear visual indication when the desired temperature has been reached.

Block Diagram



Demonstration

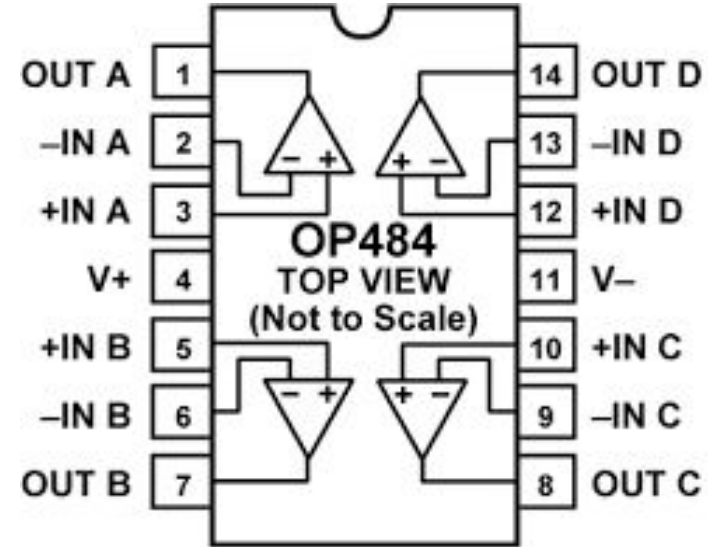
First Order Passive Filter

- The 1 μF capacitor works with the 10 $\text{k}\Omega$ resistor to form a **low-pass filter**
- This filter removes high-frequency noise from the signal coming from the first op-amp
- It smooths out rapid fluctuations in the voltage from the TMP36 sensor
- This prevents the LED from flickering due to small, unwanted signal changes
- The filtered signal is then sent to the OP97 comparator for stable switching

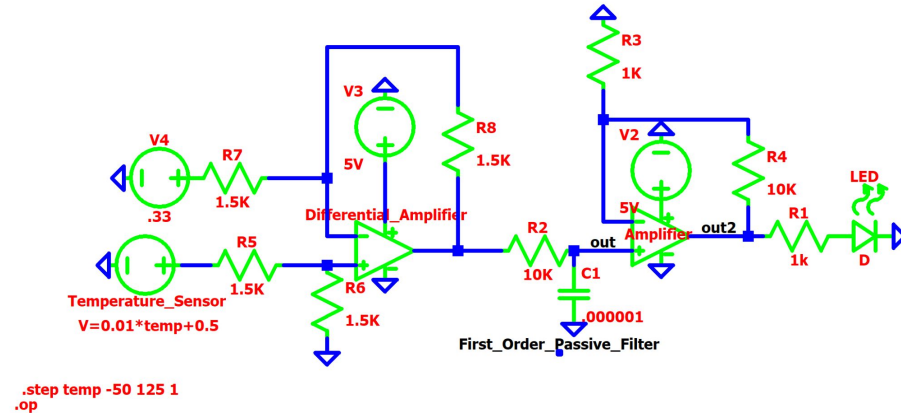
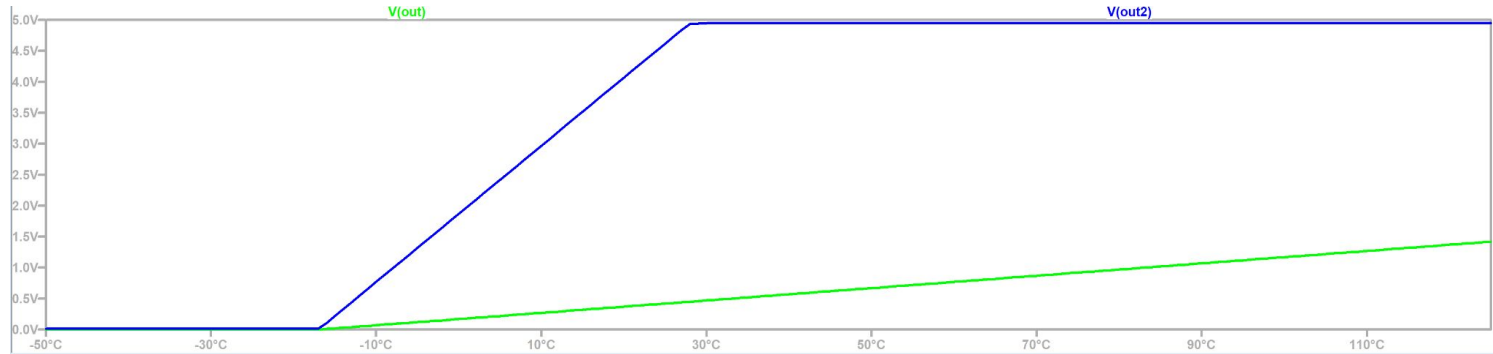


Circuit Operation Op-Amp

- The OP484 is used as a **differential amplifier**
- It takes the small voltage output from the TMP36 sensor and **outputs the difference of the Vout of the TMP36 and a threshold Vin-**
- Since our V- terminal is connected to ground and V+ is set to +5V, when the difference is negative, zero volts are outputted, and the maximum voltage is +5V
- This makes the signal easier to work with in later stages of the circuit
- The resistor network (1.5 k Ω resistors) sets the **gain of the amplifier to be 1**
- We do not yet amplify the signal because it would also amplify the noise produced by the op-amp
- The output of the OP484 is then sent to the first order passive filter for noise reduction



Simulation



Problems Encountered and Solutions

- The comparator jumping from 0 to 5V when it “turns on”
 - Decided to remove the comparator and add in a differential amplifier
- We wanted to have the comparator scale with temperature but the OP97 would not work
 - The OP97 is not a rail to rail Op-Amp which means it will go from a low state to a high state. If we wanted to get a scaling voltage with temperature, we needed a rail to rail Op-Amp. In this case we chose the OP484
- We tried to amplify the voltage in the differential voltage divider but found that it amplifies the noise created and is better to have a filter between both op amps

Design Choices

- First order passive filter:
 - We wanted to allow low frequencies through but filter out high frequencies that were creating the noise. To do this, we used the filter equation $F_c = 1/2 \cdot \pi \cdot R \cdot C$. We chose a 1uF capacitor and a 10K ohm resistor for this. This would result in a frequency cutoff of 15.9 Hz
- Amplifier Resistance
 - We chose the 1K and 10K ohm resistors as the formula for a non-inverting amplifier is $V_{out} = (1 + R_2/R_1) \cdot V_{in}$. This means that the amplifier is multiplying the voltage in by 11x. This scales until it reaches its cap of 5V because that is what is being supplied in $V+$
- OP484:
 - Is Rail to Rail to allow for a scaling of output voltage versus the low high switch of non rail to rail Op-Amps

Plans For Next Milestone(s)

For Milestone 3 we plan to add a Oscillator Circuit and a First-Order Active Filter

- An Oscillator Circuit(Astable Multivibrator) will create a continuous square wave that will allow us to control the DC motor's speed proportionally without wasting massive amounts of energy as heat.
- The First-Order Active Filter will cut down on the noise created by the motor driver. This will create a clean and sharp cutoff of power to the motor